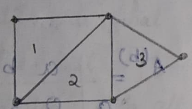
24/2/25
Monday

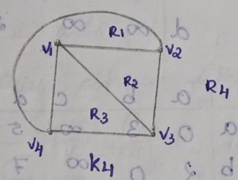
Module - 4

Planar graphs

Planar graph:

A graph G is called planar if G can be drawn in the plane with its edges intersecting only at vertices of G. [ie no intersection b/w edges].





Region of a graph:

When G [V, E] be a planar graph then a region is defined to be a area of the plane ie bounded by edges & cannot be subdivided.

→ Regions are of two types:

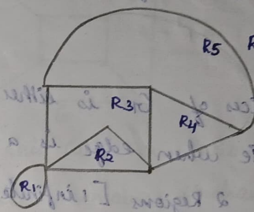
1. Finite region
2. Infinite region

1. Finite region:

If the area of the region is finite, then it is called as finite region.

2. Infinite region:

If the area of the region is infinite, then it is called an infinite region.



Here R1, R2, R3, R4, R5 are finite regions & R6 is infinite regions.

Note:

→ Only for planar graph, the regions can be found out.

→ A plannar graph contains only one infinite region.

imp EULER's THEOREM:

for xm 7 mark

Statement:

Let G = (V, E) be a connected planar graph with V vertices, E edges & R be the no. of regions of G. Then $V - E + R = 2$.

PROOF:

We prove this theorem by induction on E, the no. of edges.

Case 1: When $E = 0$

When $E = 0$, then G must have only one vertex. [Since G is connected]. & G has only one region

[Infinite region] ie we have $V=1$, $E=0$ & $R=1$.

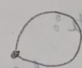
$\therefore V-E+R = 1-0+1 = 2$

That is the result is true for $E=0$.

Case 2: when $E=1$

Here the no. of vertices of G is either 1 (or a loop) or 2 [connected] ie when edge is a loop, then $V=1$, & G contains 2 Regions [1 infinite & 1 finite]

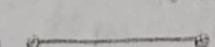
ie we have $V=1$, $E=1$, $R=2$.



$\therefore V-E+R = 1-1+2 = 2$

When E is not a loop ie E is an edge that connects two distinct vertices that is we have

$V=2$, $E=1$, $R=1$.



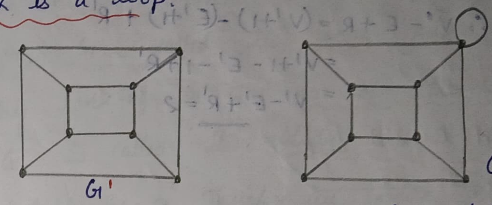
$\therefore V-E+R = 2-1+1 = 2$

Case 3: Suppose the result is true for any connected planar graph G' with k edges.

Now add a new edge k to G' to form a connected graph G. Let V', E', R' be the no. of vertices, no. of edges & no. of regions of G' respectively.

Let V, E & R be the no. of vertices, no. of edges, no. of regions of G respectively. Now, we can add the new edge k in three ways.

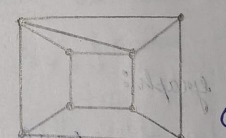
1. k is a loop.



In this case a new region is created & the no. of vertices remains unchanged ie $V=V'$, $E=E'+1$, $R=R'+1$

Then $V-E+R = V'-(E'+1)+R'+1$
$= V'-E'-1+R'+1$
$= V'-E'+R' = 2$

2. The new edge k joins two distinct vertices of G.



In this case one of the region splits, then $R=R'+1$, $E=E'+1$, $V=V'$. Therefore $V-E+R = V'-(E'+1)+R'+1$
$= V'-E'+(-1)+R'+1$
$= V'-E'+R' = 2$

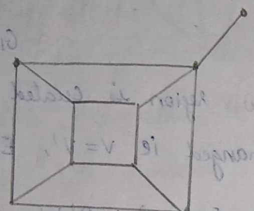
3. The new edge k is incident with only one vertex of G' & new vertex which does not belong to G'. Here a new vertex is introduced and the no.

of regions remains unchanged. $V = V'+1$, $E = E'+1$,

$R = R'$. $\therefore V - E + R = (V'+1) - (E'+1) + R'$

$= V' + \cancel{1} - E' - \cancel{1} + R'$

$= V' - E' + R' = 2$



Therefore in all the three cases the result is true

ie the result is true ~~for~~ ~~true for~~ G. By

induction it is true for all E.

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MODULE-3 :

Branches and chords of a graph:

Let T be a spanning tree of a given graph G. Then every edge of T is called a branch of ~~tree~~ T.

An edge of G that is not in a spanning tree of G is called a chord.

Note:

→ Branches & chords are defined in terms of given spanning tree.

Rank and Nullity of graphs:

The no. of Branches in a spanning tree T of a graph G is called its rank. And it is denoted by rank of G. The no. of chords of a graph G is called its nullity and it is denoted by nullity of G. Therefore we have rank of G + nullity (G) = E(G) ie no. of edges in G.

Q. Show that every graph with n vertices & E edges has (n-1) branches & (E-n+1) chords.

Ans: Let G be a graph with 'n' vertices & 'e' edges & let T be a spanning tree of G. Then, from previous theorem, we know that T has n-1 edges. Therefore no. of branches is n-1.

The no. of chords in G with respect to T is $|E(G)| - |E(T)| = e - (n-1) = e - n + 1$

Matric Definition:

A matric on a set A is a function $d: A \times A \to [0, \infty]$ where $0, \infty$ is the set of non-negative real numbers and for all x, y, z element of A, the following conditions are satisfied.

1. $d(x,y) \geq 0$ [non-negativity]

2. $d(x,y) = 0 \iff x = y$

3. $d(x,y) = d(y,x)$

4. $d(x,z) \leq d(x,y) + d(y,z)$

Prove that

Q. The distance b/w the vertices of a connected graph is a matric

Ans: PROOF:

Let U, V & W be any 3 vertices in a connected graph G, then we have

1. $d(U,V) \geq 0$

2. $d(U,V) = 0 \iff U = V$

If $U = V$, then $d(U,V) = 0$ & if $U \neq V$, $d(U,V) > 0$

3. If P is a shortest path from U to V, then P itself is a shortest path from V to U, therefore

$$d(U,V) = d(V,U)$$

4. Let ~~PUV~~ P_{UV}, P_{UW}, P_{VW} respectively be the shortest UV path, shortest UW path, & shortest VW path in the given graph G. If all UW path passes through the vertex V, then we have $P_{UW} = P_{UV} \cup P_{VW}$

Hence $d(U,W) = d(U,V) + d(V,W)$.

Otherwise P_{UW} will have smallest length than the path $P_{UV} \cup P_{VW}$. ie $d(U,W) < d(U,V) + d(V,W)$

Combining these two cases we have

$$d(U,W) \leq d(U,V) + d(V,W).$$

Hence distance b/w two vertices in a graph satisfies 3 conditions of a matric & Hence is a matric.

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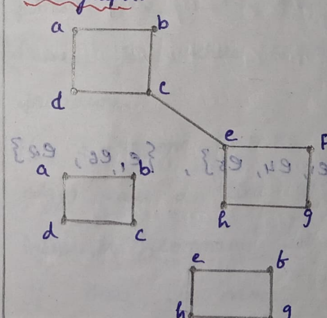
Module-4

Cut-edge:

A bridge or cut-edge or cut arc is an edge of a graph, whose deletion increases the no. of connected components.

Let G be a connected graph, and the edge $E(G)$ is called a cut edge if $G - E$ results in a disconnected graph.



By removing edge (C,E) the graph become disconnected.

∴ the edge (C,E) is the cut edge of the given graph.

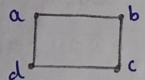
Note:

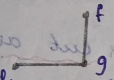
→ An edge 'e' is a cut edge if & only if the edge is not a path of any cycle in G.

→ Atleast one vertex of a cut edge is cut vertex.

→ Maximum no. of cut edges posible is $n-1$ where n is the no. of vertices.

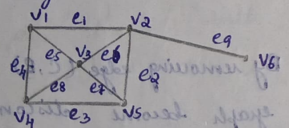
→ From the above graph clearly the vertex E is a cut vertex.





Cut set:

Let $G(V,E)$ be a ~~cut~~ connected graph a subset E' (e) is called a cut set if deletion of all edges in E' from G makes G disconnected. cut set always cuts the graph into 2.



cut sets are $\{e_9\}$, $\{e_1, e_4, e_5\}$, $\{e_1, e_6, e_2\}$ $\{e_4, e_8, e_3\}$

Edge connectivity:

Let G be a connected graph, the minimum no. of edges whose removal make graph disconnected is called edge connectivity. or ↑s the components of G.

In other words no. of edges in the smallest cutset of G is called edge connectivity of G. It is denoted by $\lambda(G)$.

If G has a 1 cut edge then $\lambda(G) = 1$

Eg: clearly the edge connectivity of above graph is 1.
ie $\lambda(G) = 1$.

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Dual graph:

Geometric dual

Consider a plannar representation of a graph with 6 regions or phases such as $R_1, R_2 \ldots R_6$. Let us place 6 points $P_1, P_2, \ldots P_6$ one on each of the regions.

Next join these 6 points according to the following procedure.

(i) If 2 regions R_i & R_j are adjacent (they have a common edge) draw a line joining that points P_i & P_j that intersects common edge b/w R_i & R_j exactly once.

(ii) If there is more than one edge common b/w

R_i & R_j draw one line b/w the points P_i & P_j for each of the common edge.

(iii) For an edge 'e' lying entirely in one region, say R_k draw a self loop at the point P_k intersecting e exactly once.

(iv) By this procedure we obtain a new graph G^*, consisting of 6 vertices $P_1, P_2 \dots P_6$ & of edges joining these vertices. Such a graph G^* is called as dual or geometric dual of G.

Properties of dual graph:

→ There is one to one correspondance b/w edges of G & its dual G^*.

→ An edge forming a self loop in G yields a pendant edge in G^*.

→ A pendant edge in G yields a self loop in G^*.

→ Edges that are in series in G, produces parallel edges in G^*.

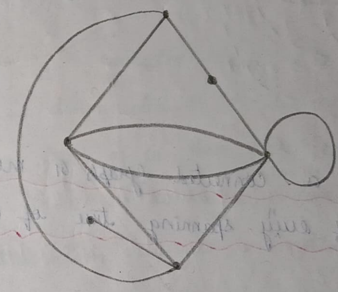
→ Parallel edges in G produces edges in series in G^*.

→ NO. of edges constituting the boundary of the region is equal to the degree of the corresponding vertex in G^* & viceversa.

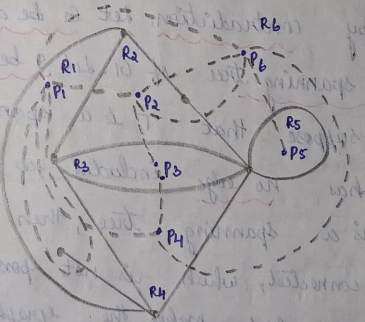
→ Graph G^* also embedded in the plane & is therefore plannar.

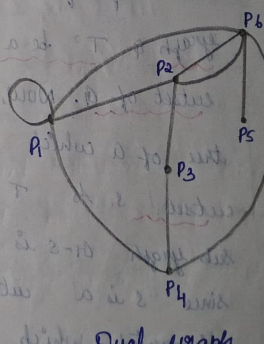
→ G is a dual of G^*, therefore instead of calling G^*, a dual of G, we usually say that G & G^* are dual graph.

1. Construct a dual graph of the following graph.



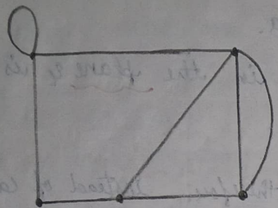
Ans:



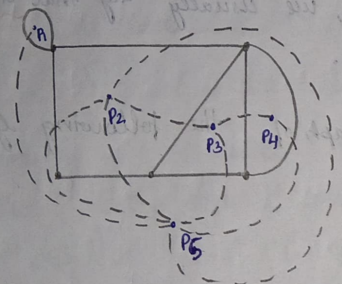


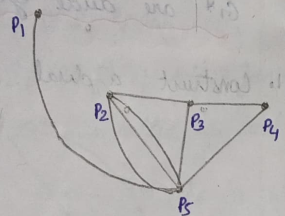
Dual graph

Q.



Ans:





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→ THEOREM :

Every cutset in a connected graph G must contain atleast one branch of every spanning tree of G.

PROOF :

The proof is by contradiction. Let G be a connected graph & T be a spanning tree of G. Let S be a cutset of G. Now, suppose that T is a spanning tree of G which has no edge included in the cutset S. As T is a spanning tree, then the sub graph G-S is connected, which is not possible since S is a cutset which makes the graph disconnected. Which is a contradiction. Hence our assumption is wrong. Therefore every cutset in a connected graph G must contain atleast one branch edge of every spanning tree.

→ THEOREM :

Let G be a connected graph & E be an edge of G, show that e is a cutedge if & only if e belongs to every spanning tree.

PROOF :

Let G be a connected graph, e be an cut edge of G ie G - e is disconnected. If 'e' does not belong to the spanning tree of G then G-e contains all the edges of that tree. G - e is connected, a contradiction. Therefore it is making a contradiction to our assumption that e is a cutedge of G. Therefore e belongs to every spanning tree.

Conversely suppose that e is in every spanning tree of G. Take e is not a cut edge. Then, G - e is connected, and thus it contains a spanning tree of G-e, which do not contain e ~~to~~ and also spanning tree of G, which is a contradiction to our assumption ~~that~~ e is in every spanning tree of G. Therefore e is a cut edge. Hence the proof.

→ THEOREM :

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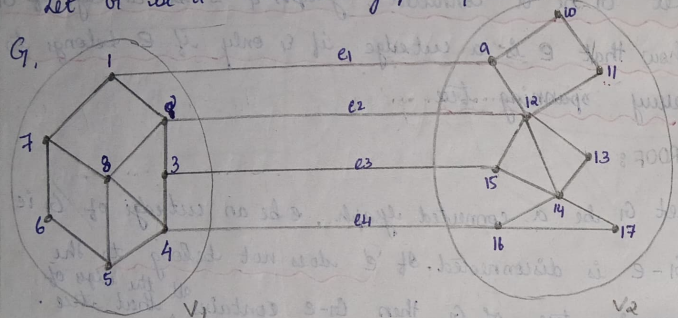
Every circuit in a graph G has an even no. of edges in common with any cutset.

PROOF :

Let G be a connected graph & s be a cutset in G.

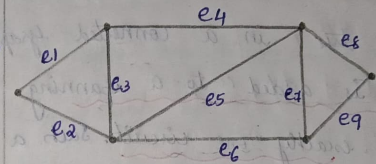


The removal of s is partitioned into 2 disjoint subsets say V_1 & V_2. ie $V = V_1 \cup V_2$ & $V_1 \cap V_2 = \phi$, null set.

Let C be a circuit in G. If all vertices of C lie entirely within the vertex set V_1 (or V_2), then the no. of edges common to C & S is zero. [ie $S \cap C = 0$] which is an even no.

If some vertices in C are in V_1 & some are in V_2, during traversing C, we need to go back & forth b/w the set of vertices V_1 & V_2. Since the circuit is closed, the no. of edges we traverse b/w V_1 & V_2 must be even. Since every edge in S has one end in V_1 & other end in V_2. & no other edge has the property of separating V_1 & V_2. Hence the no. of edges common to S & C is even.

Eg:



Consider the circuit $C = \{e_4, e_7, e_6, e_3\}$

Taking some cutsets, $S_1 = \{e_1, e_2\}$ $S_2 = \{e_6, e_7, e_9\}$

$S_3 = \{e_1, e_3, e_4\}$ $S_4 = \{e_4, e_5, e_7, e_8\}$ $S_5 = \{e_4, e_5, e_6\}$

The edges common to C & S_1 : $S_1 C = 0$,

$S_2 C = 2$, $S_3 C = 2$, $S_4 C = 2$, $S_5 C = 2$

Vertex Connectivity:

Let G be a connected graph the minimum no. of vertices whose removal makes G disconnected is called vertex connectivity of G. Vertex connectivity is denoted by $K(G)$ (kappa of G).

Fundamental cutset:

Consider a spanning tree T of a connected graph G. Take any branch b in T. Then the cutset formed by exactly one branch say b (T) & the rest of the edges in G are chords with respect to G, such as cut sets containing exactly one branch of a tree is called Fundamental Cutset.

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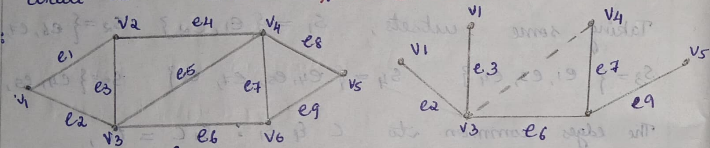
Fundamental circuit:

Let T be a spanning tree in a connected graph G, when a chord is added to a spanning tree T, then it forms exactly one circuit. Such a circuit is called a fundamental circuit.

Eg:



G.

Branch = $\{e_2, e_3, e_6, e_7, e_9\}$ Chords = $\{e_1, e_4, e_5, e_8\}$

Then we add an edge e_5 (from chord) to the spanning tree, we get a circuit (as shown above), such a circuit is called fundamental circuit.

Hence every chord of a spanning tree defines a unique fundamental circuit. Now, we write all fundamental cutsets of the given connected graph G.

Fundamental cutsets = $S_1 = \{e_2, e_1\}$, $S_2 = \{e_1, e_3, e_4\}$

$S_3 = \{e_4, e_5, e_6\}$ $S_4 = \{e_4, e_5, e_7, e_8\}$

$S_5 = \{e_8, e_9\}$

Thus the defined fundamental cutsets are unique.

THEOREM:

For any graph, the vertex connectivity $\leq$ edge connectivity $\leq$ minimum degree of a vertex of G.

If vertex connectivity is denoted as $K(G)$ edge connectivity as $\lambda(G)$ & the minimum degree of a vertex is δ, then we have to show that

$$K(G) \leq \lambda(G) \leq \delta(G) \left(\frac{2e}{n}\right)$$

PROOF:

Let G be a connected graph. First we prove the 2nd part $\lambda(G) \leq \delta(G)$.

Let v be a vertex of G, with smallest degree ie the degree of v, $v \in V = \delta$. $(d(v) = \delta)$

$\therefore$ δ edges are that incident with the vertex v are to be removed from G to isolate the vertex.

Thus, the removal of 'δ' edges disconnects the graph.

It is observed that deleting all the neighbours of a fixed vertex of minimum degree also disconnects G. And deleting all the edges that have a fixed vertex of minimum degree as an end point also disconnects G. Therefore edge connectivity $\leq$ minimum $\deg(v)$. ie $\lambda(G) \leq \delta(G)$ $\longrightarrow$ (1)

Next, to prove $\kappa(G) \le \lambda(G)$.

Let $\lambda(G)$ denote the edge connectivity & $\kappa(G)$ denote the vertex connectivity of G. Then there exists a cutset S(G) containing λ edges of G, suppose S partition's the set of vertices of G into disjoint subsets V_1 & V_2. Atmost λ edge endpoints of the edges in S lies in each of V_1 & V_2. Thus by removing atmost λ vertices from either V_1 or V_2 that are incident with the edge in S. All the edges in S will be removed from G, leaving it disconnected. ie

$\kappa(G) \le \lambda(G)$. → ②

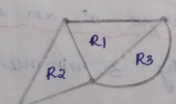
Combining eq ① & ② we get $\kappa(G) \le \lambda(G) \le \delta(G)$.

Different representation of planar graph:

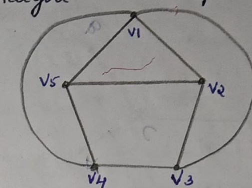
1. Plane representation
2. Straight line representation
3. Embedding on a sphere representation

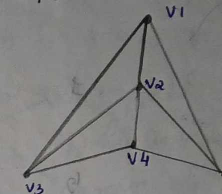
1. Plane representation:

It divides the plane into 2 regions - finite regions & infinite region.



2. Straight line representation:





Here every edge is a straight line representation.

3. Embedding on a sphere representation:

To eliminate distinction b/w finite and infinite region, a planar graph is often embedded in the surface of a sphere.

→ Put sphere on the plane & call the point of contact with plane as SP.

→ At the point SP, draw a straight line ⊥ to the plane & let the point where the line intersects the surface of the sphere NP [North pole].

→ Corresponding to any point P on the plane, there exists a unique point P' on the sphere.

→ P' is the point at which the straight line from P to the point NP intersects the surface of the sphere.

→ There is 1-1 correspondance b/w the points on the sphere & infinite points on the plane.

→ Points at ∞ in the plane corresponds to a point NP on the sphere.

→ The following fig represents the stereographic projection of planar graph on a sphere: